Student Patent Novelty Check

Provide more detail → better retrieval. Use keywords and filters to steer the search.

API: `http://127.0.0.1:8000`

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

Backend health

Search summary

Copy this into your notes or for debugging.

Copy

Mode	patentsview_live_semantic_anchor_gate (strict_anchors+keywords)
Problem	We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously…
Novelty	Uses a multi-modal transformer combining wearable time-series + patient self-reports + medication adherence to predict a…
Technologies	Cloud, IoT, Mobile, AI/ML, NLP
Keywords	heart failure, readmission, wearable, remote patient monitoring, decompensation, SpO2, respiration rate, heart rate variability, multimodal, early warning
Exclude	pacemaker, defibrillator, stent, imaging, ultrasound, catheter, printer, messaging
Assignee	Apple

Describe your idea

More input usually improves relevance — especially keywords + exclusions.

What problem are you solving?

We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously monitors vitals (heart rate variability, respiration, SpO2, activity), detects early decompensation using ML, and triggers personalized interventions. The system should work with noisy real-world signals, handle missing data, and provide explanations for alerts to clinicians.

What does your idea do?

- Automates process
- Analyzes data

- [ ] Reduces time
- [ ] Improves UX
- [ ] Hardware control

Primary domain

MedTech

Tip: Domain doesn't change semantic similarity much; keywords help more.

Key technologies

- [ ] AI/ML
- [ ] Cloud
- [ ] Mobile
- [ ] IoT
- [ ] Networking
- [ ] NLP
- [ ] Database

What's novel about your idea? (optional)

Uses a multi-modal transformer combining wearable time-series + patient self-reports + medication a

Power inputs Advanced

These steer the search hard. Comma or new-line separated.

Must-include keywords

heart rate variability
multimodal
early warning

Exclude keywords

catheter
printer
messaging

Assignee (optional)

Apple

Max results

10

Year from

2015

Year to

2025

Do you want results to show a short abstract preview?

Note: This will only display if the backend returns `abstract` / `abstract_snippet` in results.

Find Similar Patents

Results

CPC used

G16H

Idea about: We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously monitors vitals (heart rate variability, respiration, SpO2, activity), detects early decompensation using ML, and triggers personalized interventions. The system should work with noisy real-world signals, handle missing data, and provide explanations for alerts to clinicians. · Domain: MedTech · Mode: patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords) | 10 results | 1 CPC | Top match: 54%

Sort Relevance

Min similarity 0%

Reset

#1

Heartrate tracking techniques

US10874313 · 2020 · Apple Inc.

A61B | Low confidence

Similarity **54%** Top match

Show abstract preview

Embodiments of the present disclosure can provide systems, methods, and computer-readable medium for tracking the heartrate of a user using different techniques associated with different contexts. For example, motion of…

Hide why similar

Also automates part of the workflow you described.

Uses data analysis to reach decisions, similar to your idea.

Falls under a similar domain (MedTech) in the CPC scheme.

Mentions technologies that overlap with yours: Cloud, IoT, Mobile, AI/ML, NLP

Classified in CPC area A61B, which is related to your idea's focus.

Feedback

Helps us learn what "relevant" means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#2

Detecting conditions using heart rate sensors

US10524735 · 2020 · Apple Inc.

Similarity **53%**

A61B **Low confidence**

Show abstract preview

This relates to methods for measuring irregularities in a signal and corresponding devices. The devices can include a PPG sensor unit configured to detect multiple occurrences of a given event in the measured signal(s) o…

Show why similar

Feedback

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👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#3

Detecting conditions using heart rate sensors

US11793467 · 2023 · Apple Inc.

Similarity **53%**

A61B Low confidence

Show abstract preview

This relates to methods for measuring irregularities in a signal and corresponding devices. The devices can include a PPG sensor unit configured to detect multiple occurrences of a given event in the measured signal(s) o…

Show why similar

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#4

Heartrate tracking techniques

Similarity **48%**

US11690522 · 2023 · Apple Inc.

A61B Low confidence

Show abstract preview

Techniques are provided for tracking heartrate metrics using different operating modes associated with different contexts of a wearable device. For example, a heartrate sensor of the wearable device may be operated in a…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#5

Systems and methods of providing patient apps

US10733266 · 2020 · Apple Inc.

Similarity
47%

G16H Low confidence

Show abstract preview

Methods of selection of health related applications and treatment and diagnosis using health related applications are provided herein. Methods include selection of a first set of health related apps by a physician for us…

Show why similar

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#6

Methods of treatment and diagnosis using enhanced patient-physician communication

US11728030 · 2023 · Apple Inc.

Similarity
46%

G16H Low confidence

Show abstract preview

Methods of treatment and diagnosis using facilitated communication of health information from a patient device to a physician device are provided herein. Methods include improved dissemination and management of health in…

Show why similar

Feedback

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👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#7

Systems and methods for facilitating health research using a personal wearable device with research mode

Similarity **46%**

US9996678 · 2018 · Apple Inc.

G16H Low confidence

Show abstract preview

Methods and systems for facilitating health research by utilizing one or more wearable sensor devices with a research mode are provided herein. Systems include a wearable sensor device that can pair with a first portable…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#8

Systems and methods for facilitating health research using a personal wearable device with research mode

Similarity **46%**

US10460834 · 2019 · Apple Inc.

G16H Low confidence

Show abstract preview

Methods and systems for facilitating health research by utilizing one or more wearable sensor devices with a research mode are provided herein. Systems include a wearable sensor device that can pair with a first portable…

Show why similar

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👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

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#9

Systems and methods for facilitating health research using a personal wearable device with research mode

Similarity **46%**

US11056217 · 2021 · Apple Inc.

G16H **Low confidence**

Show abstract preview

Methods and systems for facilitating health research by utilizing one or more wearable sensor devices with a research mode are provided herein. Systems include a wearable sensor device that can pair with a first portable…

Show why similar

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👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

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#10

Heartrate tracking techniques

US12239427 · 2025 · Apple Inc.

Similarity

44%

A61B | **Low confidence**

Show abstract preview

An example technique may include tracking motion of a user wearing the wearable device using at least first sensors of one or more sensors of the wearable device. The technique may also include tracking a physical state…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

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Tip: If you’re getting “printer / messaging” type patents, add those words to **Exclude keywords**.